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Prompt Engineering Project-1

1. Zero-Shot Learning

Temperature = 0

You are a data extraction system.

Extract structured information from the input text and return ONLY valid JSON.

Rules:

- Output must strictly follow this schema:

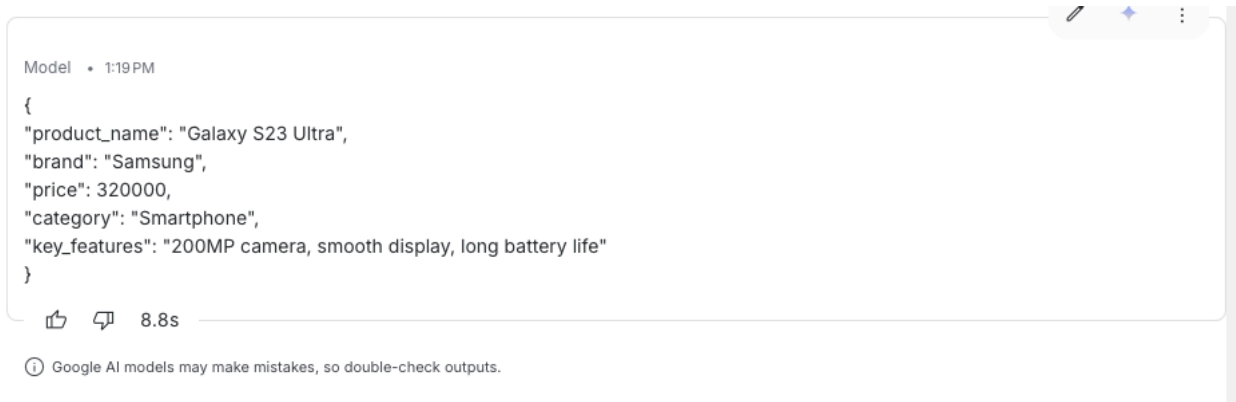
```
{
  "product_name": "",
  "brand": "",
  "price": null,
  "category": "",
  "key_features": ""
}
```
- Use "null" if information is missing
- Do not guess missing values
- Output only JSON (no explanations, no markdown, no extra text)
 - Normalize product category and features
- Keep key_features short and clean

""""

I recently bought the Samsung Galaxy S23 Ultra from a shop in Lahore for around 320,000 rupees. It has an amazing 200MP camera, super smooth display, and the battery lasts all day. Honestly, Samsung has improved a lot in this model compared to the previous ones.

""""

Model Output:



2. Few-Shot Learning

Temperature=0

You are a data extraction system.

Extract structured product information from messy, unstructured text and return ONLY valid JSON.

Rules:

- Output must strictly follow this schema:

```
{
  "product_name": "",
  "brand": "",
  "price": null,
  "category": "",
  "key_features": ""
}
```
- If information is missing, use "null."
- Keep key_features short and meaningful
- No explanations or extra text
- No markdown

Example1:

Input:

""

Just upgraded to the Google Pixel 8 Pro last week after using Pixel 6 for years. Got it for around 380,000 PKR. The camera processing is insane, especially in low light, and the AI

features like call screening are super useful. Google really improved battery efficiency too.

"""

Output:

```
{
  "product_name": "Pixel 8 Pro",
  "brand": "Google",
  "price": 380000,
  "category": "smartphone",
  "key_features": "advanced camera, low-light photography, AI features, improved battery life"
}
```

Example 2:

Input:

"""

My cousin bought the Sony PlayStation 5 from Dubai and brought it here to Pakistan. It cost him about 170,000 rupees including shipping. The graphics are super realistic and load times are almost instant. He mainly plays FIFA and Spider-Man on it all day.

"""

Output:

```
{
  "product_name": "PlayStation 5",
  "brand": "Sony",
  "price": 170000,
  "category": "gaming console",
  "key_features": "high-performance graphics, fast loading, immersive gaming experience"
}
```

Example 3:

Input:

"""

I was walking through Lahore Liberty Market and saw this Zara winter jacket. The price tag said something like 18000 but I think they were giving discount. It looked really premium, warm inside, and good for cold weather trips.

"""

Output:

```
{  
  "product_name": "winter jacket",  
  "brand": "Zara",  
  "price": 18000,  
  "category": "clothing",  
  "key_features": "warm, premium look, suitable for cold weather"  
}
```

""

I bought a OnePlus phone last week from a shop in Lahore for around 165000 rupees. It is really fast and the camera is quite good especially in night mode. Battery lasts almost the whole day even with heavy use. I don't remember the exact model name, but it's one of the newer ones. Overall I am happy with it.

""

Model Output:

```
{  
  "product_name": "OnePlus phone",  
  "brand": "OnePlus",  
  "price": 165000,  
  "category": "smartphone",  
  "key_features": "fast performance, night mode camera, long battery life"  
},
```

""

Yesterday I saw two things in a store: the Xiaomi Redmi Note 12 which was about 45000 PKR and the Realme Buds Air 3. The phone looked decent for its price and the earbuds had good sound quality. Nothing too fancy but good for daily use.

""

Model Output:

```
{
  "product_name": "Redmi Note 12",
  "brand": "Xiaomi",
  "price": 45000,
  "category": "smartphone",
  "key_features": "decent value, good for daily use"
},
{
  "product_name": "Buds Air 3",
  "brand": "Realme",
  "price": null,
  "category": "earbuds",
  "key_features": "good sound quality"
}
```

3- Edge Cases: If information is missing, the model must return null instead of guessing.

4. Justification:

Few-shot examples are used to help the model learn the correct extraction pattern from given inputs and outputs.

5. Evaluation:

- ❖ The model is tested on both clean and messy inputs.
- ❖ It almost correctly extracts structured data even in noisy text.
- ❖ It handles missing information using null values.

- ❖ Performance is validated using multiple edge-case inputs, including ambiguity and incomplete product information.